

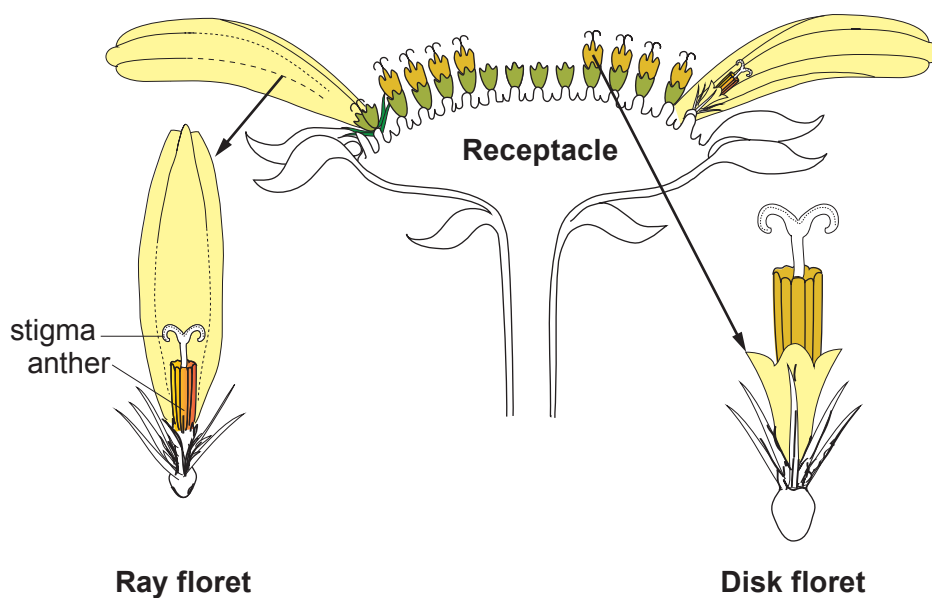
Answer all questions.

Section A

1. The photograph below shows one species of the genus *Senecio*, commonly known as groundsel. It is a plant that colonises open ground, crevices in walls and between paving slabs.



The flower head consists of many small modified flowers called florets which are cross pollinated. The structure of the flower and florets is shown.



(a) (i) State whether the florets shown in the diagram are genetically identical or genetically different. Explain your answer. [2]

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(ii) State **one** way shown in the diagram in which the structure of a floret reduces the chance of self-pollination. [1]

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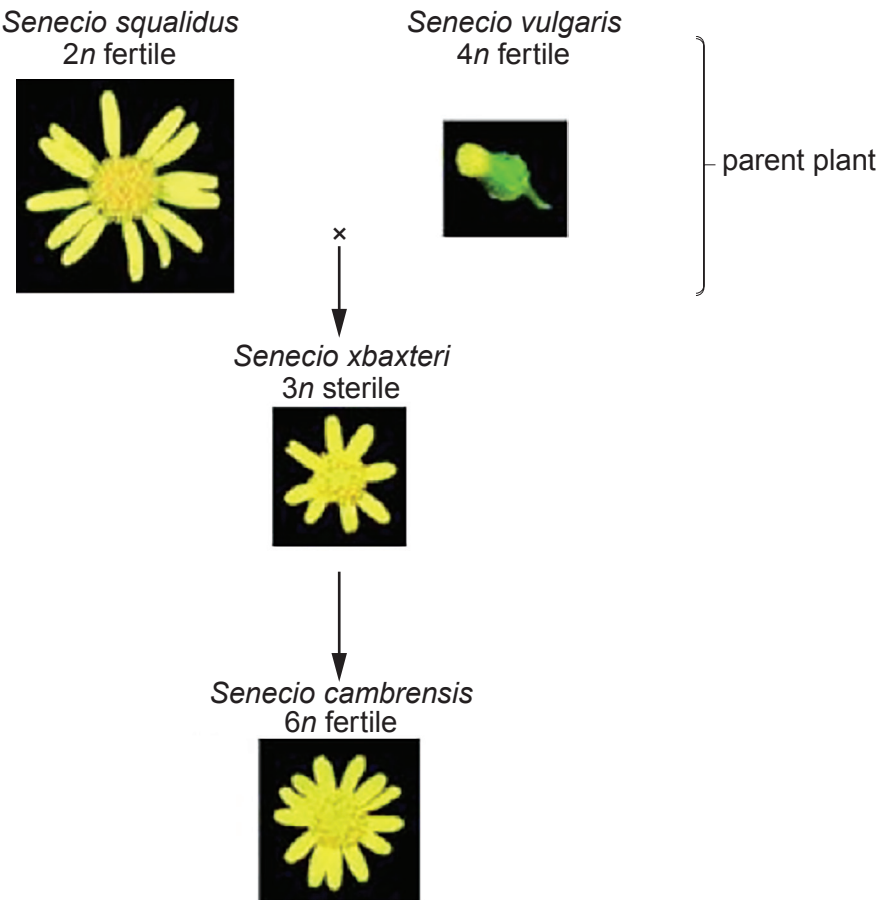
(iii) Explain the advantage of preventing self-pollination. [2]

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(b) In 1690, *Senecio squalidus* was introduced to the Oxford Botanic Garden from Italy. It escaped into the wild and became an invasive species. In North Wales, it hybridised with a native species, *Senecio vulgaris*, and gave rise to a new species, *Senecio cambrensis*, as shown below.



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(i) Leaf cells of *S.squalidus* contain 20 chromosomes. For this plant, state the number of chromosomes found in the following: [2]

I. Female gamete

II. Primary endosperm nucleus

III. Petal cells

(ii) *S.xbaxteri* has three copies of each chromosome. Describe the change that has resulted in the formation of the hybrid *S.cambrensis* and explain why it is fertile. [5]

(iii) *S. cambrensis* appeared independently in Edinburgh in 1974. Originally there were 102 plants in three populations surrounded by parental plants growing on derelict land. Suggest **one** reason why the species became extinct in Edinburgh by 1993.

[1]

